

Consider a sequence $\mathbf{x} = (x_1, x_2, \dots, x_L)$. Its type (information theory terminology) is the equivalence class of all sequences sharing the same token frequency vector f . For example, with $L = 4$ and tokens $\{0, 1\}$, the sequences $(0, 0, 1, 1)$, $(0, 1, 0, 1)$, $(1, 0, 0, 1)$, etc. all belong to type $f = (1/2, 1/2)$.

The counting factor: The number of distinct sequences in a type is the multinomial coefficient:

$$M(f) = \frac{L!}{\prod_j (L \cdot f_j)!}$$

For i.i.d. processes, every sequence in a type has the same probability $P(\mathbf{x}) = \prod_j p_j^{L f_j}$, so the total probability of the type is exactly:

$$P(\text{type } f) = M(f) \cdot P(\mathbf{x})$$

and $\log P(\mathbf{x}) + \log M(f) = \log P(\text{type } f)$ exactly.

For HMMs, sequences within a type have different probabilities (ordering matters due to transitions). So:

$$P(\text{type } f) = \sum_{\mathbf{x}: \text{freq}(\mathbf{x})=f} P(\mathbf{x}) = M(f) \cdot \mathbb{E}_{\text{uniform}}[P(\mathbf{x}) \mid \text{type } f]$$

What we compute, $\log P(\mathbf{x}) + \log M(f)$, is not exactly $\log P(\text{type } f)$. The gap is Jensen's inequality:

$$\log \mathbb{E}[P(\mathbf{x}) \mid \text{type}] \geq \mathbb{E}[\log P(\mathbf{x}) \mid \text{type}]$$

But for large L , the within-type variance of $\log P(\mathbf{x})$ shrinks (ordering becomes a lower-order effect), Jensen's gap vanishes, and our per-sequence approximation $\log P(\mathbf{x}) + \log M(f)$ converges to the true $\log P(\text{type } f)$.

That's why the type-level J matrix already matches Σ_{emp}^{-1} to $\sim 5\%$ at $L = 15$ — the Gaussian structure of the type distribution is already visible, even though the per-sequence ordering noise is still large ($R^2 = 0.6$).